

A Structured Dataset of Lufwanyama Town Council Development Projects, Budgets, Revenue and Grants

CSC 4792 — Data Mining and Warehousing
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Abstract

This dataset provides structured information relating to Lufwanyama Town Council in Zambia. It was developed as part of the CSC 4792 Data Mining and Warehousing course at the University of Zambia. The objective was to collect, extract, clean and organize publicly available local government information into structured datasets suitable for data mining and exploratory analysis.

Information was collected from the official Lufwanyama Town Council website, including council web pages, newsletters, budgets, financial documents, reports, community engagement documents and Constituency Development Fund (CDF) publications. A total of 3 official web pages and 54 PDF documents were collected. The resulting datasets cover CDF projects, council budget allocations, government grants, local revenue and selected development programmes.

1. Specifications Table

Item	Description
Subject area	Data Mining and Data Warehousing
Specific subject area	Local government development, public finance and CDF projects
Type of data	Structured tabular data
Data format	Pipe-separated CSV
Geographical coverage	Lufwanyama District, Zambia
Main source	Official Lufwanyama Town Council publications
Web pages collected	3
PDF documents collected	54
Main datasets	CDF Projects; Budget; Revenue and Grants; Development Programmes
Tools	Python, Pandas, NumPy, Requests, BeautifulSoup, PyPDF, Matplotlib, Jupyter

2. Background and Summary

Local government institutions generate information through budgets, development projects, financial statements, public notices, community engagement activities and other administrative processes. Much of this information is published in web pages and PDF documents, making direct analysis difficult. This project applies data mining and data warehousing techniques to publicly available Lufwanyama Town Council information and transforms heterogeneous source material into structured datasets.

3. Data Collection

Python was used to collect information from the council website. Requests was used to retrieve web pages while BeautifulSoup was used to process HTML content and identify relevant documents. The collection process resulted in 3 web pages and 54 PDF documents. Original source documents were retained in the data/raw/ directory for traceability.

4. PDF Text Extraction

Collected PDF documents were processed using Python and PDF text extraction. Extracted text was examined for terms associated with CDF, budgets, projects, revenue, grants, bursaries, loans, procurement, financial reporting, wards, contracts, roads, water, health and education. Relevant information was then structured into datasets.

5. Data Description

The project produces four main datasets: (1) CDF Projects, containing project identifiers, council, country, category, project amount and source information; (2) Budget, containing selected council budget indicators and allocations; (3) Revenue and Grants, containing government funding and local revenue categories; and (4) Development

Programmes, containing selected activities such as Cash for Work, farming inputs, feeder road grading, borehole drilling and CDF programmes.

6. Data Preprocessing

Preprocessing included duplicate removal, conversion of monetary values into numeric form, missing-value identification, project categorization and creation of unique project identifiers. Monetary values such as K540,195.89 were converted to numeric values such as 540195.89. Project descriptions were classified into broader categories including Education, Health, Water and Sanitation, Roads and Transport, Energy and Housing.

7. Exploratory Data Analysis

Exploratory analysis was performed using Pandas and Matplotlib. The notebook examines project counts, total and average project funding, funding by category, revenue and grant comparisons and development programme information. Visualizations provide a graphical view of development priorities and funding patterns.

8. Data Quality and Validation

Validation included checking collected documents, inspecting extracted text, identifying relevant records, removing duplicate projects, checking monetary values, assigning unique project identifiers and checking missing values. The final validation confirmed four final CSV datasets using the required pipe separator.

9. Dataset Organization

The project contains data/raw/ for original documents, data/cleaned/ for processed information, data/final/ for final CSV datasets, notebooks/ for the Jupyter Notebook, scripts/ for the collection script, and documentation/ for project documentation.

10. Reproducibility

The main analysis notebook is lufwanyama_data_mining.ipynb. The collection script is collect_lufwanyama.py. The workflow covers document collection, PDF extraction, information identification, data cleaning, structuring, exploratory analysis, visualization and CSV export. Final CSV files use the pipe character (|) as the field separator.

11. Intended Uses

The dataset can support analysis of local government development spending, CDF projects, public finance, revenue and grants, development-sector comparisons, funding patterns, development planning research, data visualization and educational data mining.

12. Limitations

The dataset is based on publicly available council information. Information that was not publicly available could not be included. Some PDF documents contain scanned or image-based pages and may therefore provide limited automatically extracted text. Source documents also vary in structure and completeness. The dataset represents selected structured information rather than every item contained in every collected document.

13. Ethical Considerations

The dataset was developed using publicly available council information and does not intentionally collect private personal information. It is intended for academic and analytical purposes. Users should maintain appropriate attribution to the original sources and avoid misrepresenting council publications.

14. Conclusion

This project demonstrates how publicly available local government information can be transformed from web pages and PDF documents into structured datasets suitable for data mining and analysis. The resulting datasets provide information about CDF projects, council budgets, government grants, local revenue and development programmes and provide a foundation for further research into local government development in Lufwanyama District.

15. References

Lufwanyama Town Council. Official council website and publicly available council publications.
Lufwanyama Town Council. Constituency Development Fund publications.
Lufwanyama Town Council. Approved budget publications.
Lufwanyama Town Council. Financial and performance reports.
Lufwanyama Town Council.

Council newsletters and community engagement publications.
University of Zambia. CSC 4792 Data Mining and Warehousing project requirements.